

Regional Income Growth and House Price Dynamics in the Netherlands

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2026-06-10

Set-up your environment

Regional Income Growth and House Price Dynamics in the Netherlands

Tutorial group 3.3

Part 1 - Identify a Social Problem

Use APA referencing throughout your document. Here's a link to some explanation.

1.1 Describe the Social Problem

The Dutch housing market has been under pressure for several years now. Between 2013 and 2023 the average price of houses has more than doubled, while the income growth in the same period considerably lagged behind. The increasing gap between housing costs and purchasing power is not only an economic phenomenon, but also a social problem; it determines who does have access to a stable living situation and who doesn't.

It's striking that this development differs significantly across regions in the Netherlands. In the Randstad, prices increased much more than in the peripheral regions like Groningen or Zeeland, while the income development per region differs significantly too. Factors such as access to big employers, education level and local economic structure play an important role in this shift. And it is because of this regional variety that it's interesting to investigate the relation between income and house prices.

This project investigates the relationship between regional income growth and house price dynamics in Dutch provinces, based on CBS-data and the programming tool R. The goal is to discover in which regions the housing market detached most from the local income development.

Part 2 - Data Sourcing

2.1 Load in the data

Preferably from a URL, but if not, make sure to download the data and store it in a shared location that you can load the data in from. Do not store the data in a folder you include in the Github repository!

joining

2.2 Provide a short summary of the dataset(s)

In this case we see 28 variables, but we miss some information on what units they are in. We also don't know anything about the year/moment in which this data has been captured.

These are things that are usually included in the metadata of the dataset. For your project, you need to provide us with the information from your metadata that we need to understand your dataset of choice.

2.3 Describe the type of variables included

Think of things like:

- Do the variables contain health information or SES information?
- Have they been measured by interviewing individuals or is the data coming from administrative sources?

Part 3 - Quantifying

3.1 Data cleaning

Data cleaning coding datacombined15tot23goed.

this is how we cleaned this data set:

Data cleaning coding woningprijsprov: this is how we cleaned this dataset:

Please use a separate 'R block' of code for each type of cleaning. So, e.g. one for missing values, a new one for removing unnecessary variables etc.

3.2 Generate necessary variables

Variable 1 maken van nieuwe variabele 1, we hebben hier het gemiddelde besteedbare inkomen van iedere provincie over 8 jaar. aldus het gemiddeld

```
gem_inkomen_provincie <- aggregate(gem_bstb_inkomen_x_100 ~ Provinces,
                                   data = datacombined15tot23goed,
                                   FUN = mean)

write.csv(gem_inkomen_provincie, "goede data/gem_inkomen_provincie.csv", row.names = FALSE)

colnames(gem_inkomen_provincie)[colnames(gem_inkomen_provincie) == "gem_bstb_inkomen_x_100"] <- "gem_bs

write.csv(gem_inkomen_provincie, "goede data/gem_inkomen_provincie.csv", row.names = FALSE)
```

Variable 2 maken van nieuwe variabele 2 gebruik gemaakt van de oorspronkelijke dataset:datacombined15tot23goed. waarin we de gewogen gemiddeldes van het inkomen van verschillende provincies: binnen en buiten de randstad hebben gerepresenteerd

de vorm van de huidige gecleande dataset aanpassen om dan zo verder te kunnen werken met een vernieuwde dataset en ook een vernieuwde dataset. omdat we voor deze nieuwe variabele eigenlijk onze twee datasets moeten samenvoegen.

```
woningprijsprov <- read.csv("goede data/goedehuizenprijzenperprov.csv", check.names = FALSE)

woningprijsprov <- woningprijsprov[!grepl("Bron", woningprijsprov$`Regio.s`), ]

woningprijsprov$Onderwerp <- NULL
woningprijsprov$`...4` <- NULL

woningprijsprov_lang <- reshape(woningprijsprov,
                                varying = c("2015", "2020", "2021", "2022", "2023"),
                                v.names = "gem_huizenprijs",
                                timevar = "jaar",
                                times = c(2015, 2020, 2021, 2022, 2023),
                                direction = "long")

woningprijsprov_lang$id <- NULL

write.csv(woningprijsprov_lang, "goede data/goedehuizenprijzenperprov_lang.csv", row.names = FALSE)
```

3.3 Visualize temporal variation

3.4 Visualize spatial variation

Here you provide a description of why the plot above is relevant to your specific social problem.

3.5 Visualize sub-population variation

What is the poverty rate by state?

Here you provide a description of why the plot above is relevant to your specific social problem.

3.6 Event analysis

Analyze the relationship between two variables.

Here you provide a description of why the plot above is relevant to your specific social problem.

Part 4 - Discussion

4.1 Discuss your findings

Part 5 - Reproducibility

5.1 Github repository link

Provide the to your PUBLIC repository here: ...

5.2 Reference list

Use APA referencing throughout your document.